

Section 2.8: Subspaces of $\mathbb{R}^n$

Section 2.9: Dimension and Rank

Subspaces of $\mathbb{R}^n$

- **Definition:** A **subspace** of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:
 - a) The zero vector is in H .
 - b) For each $\mathbf{u}$ and $\mathbf{v}$ in H , the sum $\mathbf{u} + \mathbf{v}$ is in H .
 - c) For each $\mathbf{u}$ in H and each scalar c , the vector $c\mathbf{u}$ is in H .

Subspaces of $\mathbb{R}^n$

- **Example:** If $\mathbf{v}_1$ and $\mathbf{v}_2$ are in $\mathbb{R}^n$ and $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, then H is a subspace of $\mathbb{R}^n$.

Proof:

- a. Note that the zero vector is in H because $\mathbf{0} = 0\mathbf{v}_1 + 0\mathbf{v}_2$.
- b. Take two arbitrary vectors in H , say,

$$\mathbf{u} = s_1\mathbf{v}_1 + s_2\mathbf{v}_2 \quad \text{and} \quad \mathbf{v} = t_1\mathbf{v}_1 + t_2\mathbf{v}_2.$$

Then $\mathbf{u} + \mathbf{v} = (s_1 + t_1)\mathbf{v}_1 + (s_2 + t_2)\mathbf{v}_2$, which is in H .

- c. Also, for any scalar c , the vector $c\mathbf{u}$ is in H , because $c\mathbf{u} = c(s_1\mathbf{v}_1 + s_2\mathbf{v}_2) = cs_1(\mathbf{v}_1) + cs_2(\mathbf{v}_2)$.

Column Space and Null Space of A Matrix

- **Definition:** The **column space** of an $m \times n$ matrix A is the set $\text{Col } A$ of all linear combinations of the columns of A .
- If $A = [\mathbf{a}_1 \cdots \mathbf{a}_n]$, with columns in $\mathbb{R}^m$, then

$$\text{Col } A = \text{Span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}.$$

- The column space of an $m \times n$ matrix is a subspace of $\mathbb{R}^m$.

Column Space and Null Space of A Matrix

- **Example:** Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$.

Determine whether $\mathbf{b}$ is in the column space of A .

- **Solution:** Row reducing the augmented matrix $[A \ \mathbf{b}]$,

$$\begin{bmatrix} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & -2 & -6 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

We can conclude that $A\mathbf{x} = \mathbf{b}$ is consistent and $\mathbf{b}$ is in $\text{Col } A$.

Column Space and Null Space of A Matrix

- **Definition:** The **null space** of a matrix A is the set $\text{Nul } A$ of all solutions of the homogenous equation $A\mathbf{x} = \mathbf{0}$.
- **Theorem:** The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.
- **Proof:**
 - a. The zero vector is in $\text{Nul } A$ (because $A\mathbf{0} = \mathbf{0}$).
 - b. Take any $\mathbf{u}$ and $\mathbf{v}$ in $\text{Nul } A$, then
$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0},$$
so $\mathbf{u} + \mathbf{v}$ is in $\text{Nul } A$.
 - c. For any scalar c , $A(c\mathbf{u}) = cA\mathbf{u} = c\mathbf{0} = \mathbf{0}$, so $c\mathbf{u}$ is in $\text{Nul } A$.

Basis for A Subspace

- **Definition:** A **basis** for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H .
- **Example:** The columns of an invertible $n \times n$ matrix form a basis for all of $\mathbb{R}^n$ because they are linearly independent and span $\mathbb{R}^n$, by the Invertible Matrix Theorem.

Basis for A Subspace

- One such matrix is the $n \times n$ identity matrix. Its columns are denoted by $\mathbf{e}_1, \dots, \mathbf{e}_n$:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

- The set $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is called the **standard basis** for $\mathbb{R}^n$.

Basis for A Subspace

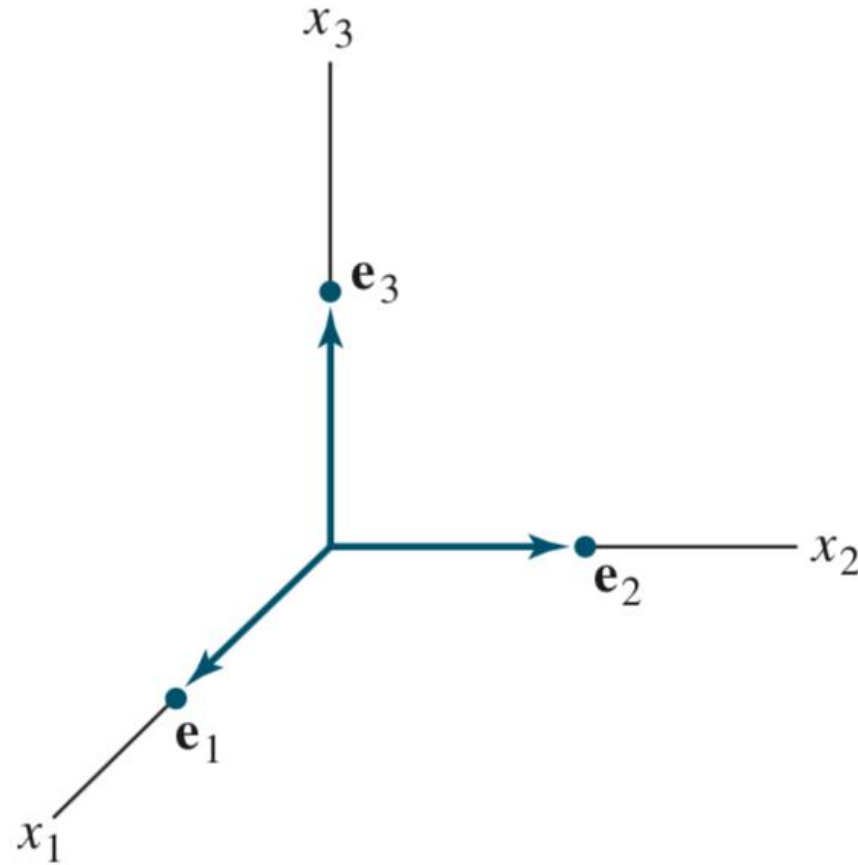


FIGURE 3

The standard basis for $\mathbb{R}^3$.

Basis for A Subspace

- **Theorem:** The pivot columns of a matrix A form a basis for the column space of A .
- **Example**

$$A = \begin{bmatrix} 2 & 4 & 2 & 6 \\ 1 & 4 & 3 & 5 \\ 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 1 \end{bmatrix} \sim \dots \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- Basis for Col $A = \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 4 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 6 \\ 5 \\ 2 \\ 1 \end{bmatrix} \right\}$ Basis for Nul $A = \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} \right\}$

Basis for A Subspace

Example: Find a basis for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Solution: First, write the solution of $A\mathbf{x} = \mathbf{0}$ in parametric vector form:

$$[A \quad \mathbf{0}] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{array}{l} x_1 - 2x_2 - x_4 + 3x_5 = 0 \\ x_3 + 2x_4 - 2x_5 = 0 \\ 0 = 0 \end{array}$$

The general solution is $x_1 = 2x_2 + x_4 - 3x_5$, $x_3 = -2x_4 + 2x_5$, with x_2 , x_4 , and x_5 free.

Basis for A Subspace

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $\mathbf{u} \quad \quad \mathbf{v} \quad \quad \mathbf{w}$

$$= x_2 \mathbf{u} + x_4 \mathbf{v} + x_5 \mathbf{w} \tag{1}$$

Equation (1) shows that $\text{Nul } A$ coincides with the set of all linear combinations of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$. That is, $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ generates $\text{Nul } A$. In fact, this construction of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ automatically makes them linearly independent, because equation (1) shows that $\mathbf{0} = x_2 \mathbf{u} + x_4 \mathbf{v} + x_5 \mathbf{w}$ only if the weights x_2 , x_4 , and x_5 are all zero. (Examine entries 2, 4, and 5 in the vector $x_2 \mathbf{u} + x_4 \mathbf{v} + x_5 \mathbf{w}$.) So $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a *basis* for $\text{Nul } A$. ■

Section 2.4: Partitioned Matrices

Section 2.5: The LU Factorization

Section 2.8: Subspaces of $\mathbb{R}^n$

Section 2.9: Dimension and Rank

Coordinate Systems

- Suppose $B = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ is a basis for H , and suppose a vector $\mathbf{x}$ in H can be generated in two ways, say,

$$\mathbf{x} = c_1\mathbf{b}_1 + \dots + c_p\mathbf{b}_p \quad \text{and} \quad \mathbf{x} = d_1\mathbf{b}_1 + \dots + d_p\mathbf{b}_p \quad (1)$$

- Then, subtracting gives

$$0 = \mathbf{x} - \mathbf{x} = (c_1 - d_1)\mathbf{b}_1 + \dots + (c_p - d_p)\mathbf{b}_p \quad (2)$$

- Since B is linearly independent, the weights in (2) must all be zero. That is, $c_j = d_j$ for $1 \leq j \leq p$, which shows that the two representations in (1) are actually the same.

Coordinate Systems

- **Definition:** Suppose the set $B = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ is a basis for a subspace H . For each $\mathbf{x}$ in H , the coordinates of $\mathbf{x}$ relative to the basis B are the weights $c_1, \dots, c_p$ such that $\mathbf{x} = c_1\mathbf{b}_1 + \dots + c_p\mathbf{b}_p$, and the vector in $\mathbb{R}^p$

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ \cdot \\ \cdot \\ \cdot \\ c_p \end{bmatrix}.$$

- is called **the coordinate vector of $\mathbf{x}$ (relative to B)** or the **B -coordinate vector of $\mathbf{x}$.**

Coordinate Systems

- **Example:** Let $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$, and $B = \{\mathbf{v}_1, \mathbf{v}_2\}$.
- Then B is a basis for $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ because $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent.
- Determine if $\mathbf{x}$ is in H , and if it is, find the coordinate vector of $\mathbf{x}$ relative to B .
- **Solution** If $\mathbf{x}$ is in H , then the following vector equation is consistent:
$$c_1 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

Coordinate Systems

- The scalars c_1 and c_2 , if they exist, are the B-coordinates of $\mathbf{x}$.
Row operations show that

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

- Thus $c_1 = 2$, $c_2 = 3$ and $[\mathbf{x}]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. The basis determines a “coordinate system” on H , which can be visualized by the grid shown on the next slide.

Coordinate Systems

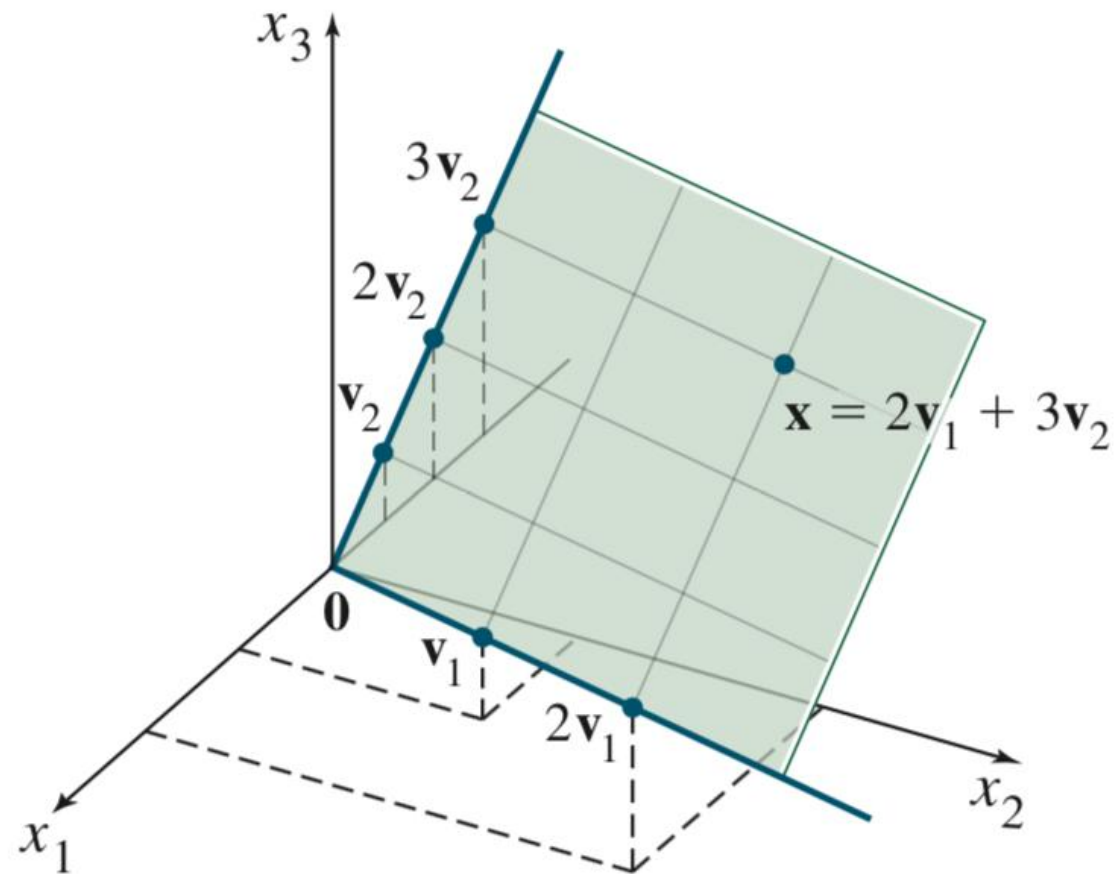


FIGURE 1 A coordinate system on a plane H in $\mathbb{R}^3$.

The Dimension of A Subspace

- **Definition:** The **dimension** of a nonzero subspace H , denoted by **$\dim H$** , is the number of vectors in any basis for H . The dimension of the zero subspace $\{0\}$ is defined to be zero.
- **Definition:** The **rank** of a matrix A , denoted by **$\text{rank } A$** , is the dimension of the column space of A .

The Dimension of A Subspace

- **Example:** Determine the rank of the matrix

$$A \sim \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & -6 & 4 & 14 & -20 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix}$$

- **Solution** Reduce A to echelon form:

$$A \sim \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & -6 & 4 & 14 & -20 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix} \sim \dots \sim \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 4 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

pivot columns 

- The matrix A has 3 pivot columns, so $\text{rank } A = 3$.

The Dimension of A Subspace

- **Theorem:** If a matrix A has n columns, then
$$\text{rank } A + \dim \text{Nul } A = n.$$

- **Theorem:** Let H be a p -dimensional subspace of $\mathbb{R}^n$.
 - a. Any linearly independent set of exactly p elements in H is automatically a basis for H .
 - b. Also, any set of p elements of H that spans H is automatically a basis for H .

Rank and the Invertible Matrix Theorem

- **The Invertible Theorem (continued)** Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.
 - m. The columns of A form a basis of $\mathbb{R}^n$.
 - n. $\text{Col } A = \mathbb{R}^n$
 - o. $\text{rank } A = n$
 - p. $\dim \text{Nul } A = 0$
 - q. $\text{Nul } A = \{\mathbf{0}\}$

Rank and the Invertible Matrix Theorem

- **Proof** Statement (m) is logically equivalent to statements (e) and (h) regarding linear independence and spanning. The other five statements are linked to the earlier ones of the theorem by the following chain of almost trivial implications:

$$(g) \Rightarrow (n) \Rightarrow (o) \Rightarrow (p) \Rightarrow (q) \Rightarrow (d)$$

- Statement (g), which says that the equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$, implies statement (n), because $\text{Col } A$ is precisely the set of all $\mathbf{b}$ such that the equation $A\mathbf{x} = \mathbf{b}$ is consistent.

Rank and the Invertible Matrix Theorem

- The implications $(n) \Rightarrow (o)$ follow from the definitions of *dimension* and *rank*.
- If the rank of A is n , the number of columns of A , then $\dim \text{Nul } A = 0$, by the Rank Theorem, and so $\text{Nul } A = \{\mathbf{0}\}$. Thus $(o) \Rightarrow (p) \Rightarrow (q)$.
- Also, statement (q) implies that the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, which is statement (d).
- Since statements (d) and (g) are already known to be equivalent to the statement that A is invertible, the proof is complete.